



PSOC™ Edge EPC 2 vs EPC 4

Customer training workshop (CTW)

April 2026



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Introduction to Edge Protect category

The Edge Protect Category (EPC) categorises products into different security levels, each representing a specific set of security features and requirements.

These levels include:

- **EPC 1:** Basic security features, such as secure boot, firmware update, and encryption
- **EPC 2:** Enhanced security features, including secure communication and secure key storage
- **EPC 3:** Advanced security features, such as hardened cryptography and fault attack detection
- **EPC 4:** High-level security features, including a hardware-isolated, payment terminal-grade, and secure enclave

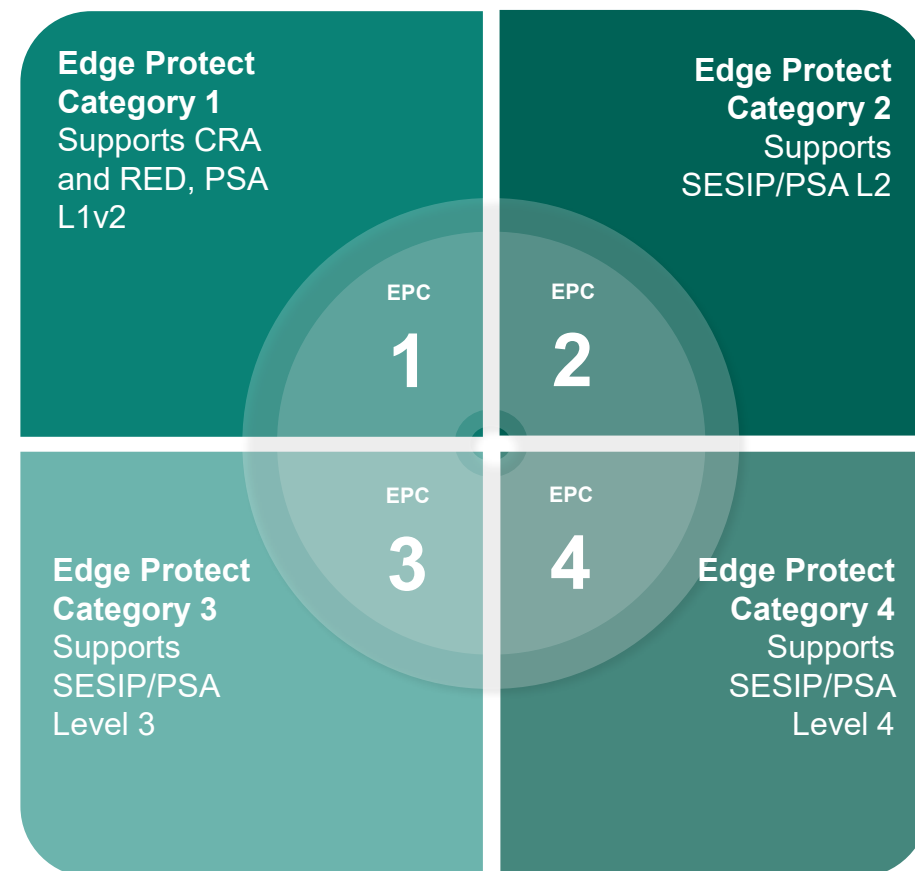


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Security features of EPC categories

Edge Protect Category (EPC) #		EPC 1	EPC 2	EPC 3	EPC 4
Definition and Compliance		Minimum ICW MCU requirements SESIP 1 + PSA L1v2	Additional wireless connectivity security services SESIP L2 + PSA L2	Hardened accelerated crypto and fault, glitch mitigation SESIP L3 + PSA L3	Hardware isolated, lockstep Secure Enclave (SE) SESIP/PSA3 + L4 iSE
Features	CRA/RED support	✓ CRA Class 1	✓ CRA Class 1	✓ CRA Class 2	✓ CRA Class 2
	Cryptography (PQC support starting '26)	✓ Accelerated	✓ Accelerated	✓ Hardened Accelerated	✓ Advanced Accelerated
	Secured boot	✓	✓	✓	✓
	Secured FW update	✓	✓	✓	✓
	Processing isolation (Trust Zone)	✓	✓	✓	✓
	Physical fault attack mitigation	✓ Boot-time	✓ Boot-time	✓ Boot-time/Run-time	✓ Boot-time/Run-time
	Prebuilt security stack available (TF-M)		✓	✓	✓
	Fault sensor management system			✓	✓
	DICE support for SPDM			✓ DICE Ready	✓
	Hardware isolated secure enclave				✓

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Security features of PSOC™ Edge EPC 2 / EPC4

Security feature	PSOC™ EPC 2	PSOC™ EPC 4
Hardware-isolated Secure Enclave	No	Yes
Secure boot and secure FW update	Yes	Yes
Secure debug	Yes	Yes
Processing isolation	Yes	Yes
Compliance with CRA / RED	Yes	Yes
Physical glitch hardening	No	Yes
DICE compliance	No	Yes
SE user key storage	No	Yes
SE RT services (CRYPTO, attestation)	No	Yes
SE RT services update	No	Yes
PSA certification compliance	Level 2	Level 4

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Key differences for EPC 2 / EPC 4

RRAM memory space

EPC 2	EPC 4
The RRAM memory available for the user is ~356 KB	The RRAM memory available for user is ~188 KB; due to SERT services stored in the RRAM

Key storage

EPC 2	EPC 4
OEM can store the keys in the RRAM using CM33 S application (PC=2)	OEM can import keys within the Secure Enclave using the SE RT services. The user keys can also be imported into the Secure Enclave via provisioning

Key differences for EPC 2 / EPC 4 (contd.)

PSA root-of-trust (RoT) keys

EPC 2	EPC 4
– The PSA root-of-trust (RoT) keys, such as initial attestation key (IAK), hardware unique key (HUK) and Root-of-trust public key (ROTPK), are protected by Secure Enclave and cannot be accessed by the EPC 2 device, as it does not have SE RT services – A copy of these keys is available in RRAM for EPC 2 and can be accessed by the CM33 S (PC=2) application	The PSA root-of-trust (RoT) keys, such as initial attestation key (IAK), hardware unique key (HUK) and Root-of-trust public key (ROTPK) are protected by Secure Enclave and can be accessed by EPC 4 device

Key differences for EPC 2 / EPC 4 (contd.)

Protection context (PC) switching

EPC 2	EPC 4
In the EPC 2 device, CM33 S and CM33 NS are in the same protection context (PC=2). It does not require any PC switching for communication between CM33 NS and CM33 S.	In the EPC 4 device, the TF-M executes in PC= 2 and configures the CM33 NS to be executed in the PC= 5. Communication between CM33 NS and CM33 S requires PC switching. This is taken care by TF-M for EPC 4 device.

Default TF-M storage

EPC 2	EPC 4
Default TF-M code on the EPC 2 device resides in the external flash memory.	On the EPC 4 parts, at device boot up, the TF-M image is copied from the external flash to internal flash for execution. The executing code from the internal flash is more secure.

Key differences for EPC 2 / EPC 4 (contd.)

Encryption support

EPC 2	EPC 4
XIP encryption is supported by Edge Protect Bootloader.	Secure Enclave based encryption mechanism is supported by Edge Protect Bootloader.

Revision history

Document revision	Date	Description of change
**	2026-04-25	Initial release.

